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Table of Contents:

Cover

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A Short Editorial

Question I: Can we build materials which can repair themselves?

Question II: Why can’t humans regenerate limbs?

Question III: What happens in Neutron Stars?

Question IV: Can we get electricity from trash?

Question V: Can anybody predict the timing of earthquakes accurately?

Acknowledgements

Conclusion

A Short Editorial:

YSJOI - by the youth, for the youth of Iraq!

In the great land that was once Mesopotamia, and the home of the hanging gardens of Babylon, progress in science has slowed.

Why?

Because the youth of Iraq have no platform to build upon! Opportunities to learn about science, technology, engineering, and mathematics are rare, and students trying to learn about them are even actively discouraged – I speak from experience.

But instead of surrendering to that silence, I have chosen to rally the youth of Iraq. If we do not have a base to stand on, we will build one!

And the first step to that is this Youth Science Journal Of Iraq, YSJOI, where we aim to provide that very base – an opportunity to teach, learn and explore through different articles – be it inspiring them, writing them, or reading them!

Thank you for being part of this beginning.
Now, dive in – into the weird, the beautiful, and the wonderful.

– Satvik M. Bhure, Founder of YSJOI

Question I: Can we build materials which can repair themselves?

– Ramyar Abdullah, 15

Creating materials that can repair themselves might seem challenging at first— is it even possible? Well, it is, and it has been done before. Some examples include self-healing concrete, rubber, and polymers (plastics).

Self-healing concrete is produced with tiny capsules that contain bacteria as well as nutrients for them. After installation, these microorganisms usually stay dormant. However, if cracks form, water will most likely enter the concrete through the crack and reach the bacteria. Once this happens, the bacteria become active and use the nutrients in the capsules to produce limestone, filling the crack.

However, there are a lot of limitations to self-healing concrete. Firstly, if the weather is very dry and no water enters the crack, the bacteria stay dormant and will not produce limestone. Secondly, the bacteria can only fill small cracks, about 1 millimeter thick. Finally, the nutrients eventually run out, after which the bacteria can no longer produce limestone.

Self-healing rubber works with reversible chemical reactions. Once the rubber is torn and the two pieces are put back together, new chemical bonds form, resulting in the rubber becoming whole again. However, this needs human assistance to put the torn pieces back together if there are large tears.

Moreover, particles surrounding the rubber might interfere with the reactions, so the surfaces need to stay clean and free of impurities such as dirt.

Self-healing polymers (plastics)

There are two types of self-healing polymers. One works with capsules that are filled with a healing material, which fills cracks through a series of chemical reactions. The other type works just like self-healing rubber, using reversible chemical bonds to repair itself.

Plastics have similar limitations to either rubber or concrete, depending on how they work.

We are still far from creating truly self-healing materials. However, science is advancing rapidly, and researchers are continuing to improve the self-healing materials we already have.

Question II: Why can't humans regenerate limbs?

– *Koya Rostam, 12*

Humans cannot grow back lost limbs due to an evolutionary trade-off prioritizing rapid survival over long-term tissue regeneration. While creatures like salamanders can regrow some limbs, mammals evolved a highly advanced immune system that aggressively fights infection but completely halts the process of limb regrowth. Growing a full limb demands a massive amount of metabolic energy that early human ancestors could not afford to waste.

When a severe injury occurs, human cells quickly form thick scar tissue to instantly stop bleeding and prevent deadly infections. However, this fast-healing scar barrier physically blocks the deep cellular signaling required to rebuild complex bone, nerve, and muscle structures.

True regeneration requires blastema formation, where specialized cells revert to stem cells to rebuild a missing limb from scratch completely. Humans lack the specific genetic triggers needed to switch adult cells back into this flexible, embryonic state. Instead, our ancestors relied on quick healing to stay mobile and escape predators.

At a cellular level, mammals have deactivated the genes responsible for large-scale tissue regrowth. This inactivation likely evolved as a defense mechanism against cancer. The same cellular signals that allow cells to divide rapidly and freely during limb regeneration also carry a high risk of causing uncontrolled tumor growth in warm-blooded animals.

Furthermore, human nerves do not regenerate efficiently over long distances. In a severed limb, the complex network of blood vessels, muscle fibers, and nerve pathways cannot realign properly without a biological blueprint. Because our bodies prioritize sealing the wound rather than maintaining this blueprint, the pathway for regrowth is permanently lost.

In summary, the human body chooses immediate survival through scarring over the slow, energy-expensive process of rebuilding complex anatomical structures.

Question III: What happens in Neutron Stars?

– Hussam Majid, 15

Neutron stars. One of the possible fates a star can have. What happens in these high-density, neutron-packed clusters of energy?

To begin with, for a star to even become a neutron star, several criteria have to be met. Firstly, the star has to be at least eight solar masses. Or, eight times the mass of our Sun. The numbers get absurd rather quickly.

When such massive stars begin their stable lives, they have mainly hydrogen and some helium in their cores, like all stars. The nuclear fusion of hydrogen produces an outward force of radiation pressure, counteracting the inward force of gravity. Eventually, the hydrogen fuel runs out. The star becomes desperate.

Not wanting to be crushed by gravity, the helium burns and carbon is made. This cycle continues, and the elements produced get heavier and heavier. Until iron is made. When that happens, energy is not released. It is absorbed.

Production of iron requires more energy than it releases. That's when gravity wins, and the star collapses. However, this collapsing is not soft. Some atoms get ripped apart into smaller units. Protons and electrons are combined in that great force to form neutrons and neutrinos. The small neutrinos are flung out into space. The core becomes a ball of neutrons, while the crust is made of iron after it cools. But gravity doesn't stop.

The core shrinks to a size where it can no longer get smaller. It becomes stiff. It fights gravity with one last stand. All the material and heavy elements that are being pushed inside the core rebound from its high density. This, along with the enormous number of neutrinos flooding out, causes the outer layers of the star to be blown away, and heavy elements are scattered into space in what is called a core-collapse supernova.

What's left is called a neutron star. The outer crust is made of heavily packed iron. The inner crust contains very dense nuclei surrounded by a sea of delocalized neutrons. The outer core is made up mainly of neutrons. The inner core?

No one knows. The temperature and pressure are far too high even to predict what is inside there. There's more. A neutron star's mass is around 1.2-2.2 solar masses. Its diameter is around 20km across. The density is ridiculous. However, as the star shrinks, its magnetic field lines get closer. This makes the magnetic field trillions of times stronger than our own Earth's. To conclude, neutron stars are a fascinating phenomenon of space that we can't fully explain. It blends theoretical, quantum, and nuclear physics into one topic that baffles many scientists.

Question IV: Can we get electricity from trash?

– Ibrahim Omar, 17

Every day, millions of people throw away rubbish. Household waste which eventually ends up in huge landfill sites, where it sits piled up for a very long time. As cities continue to expand and the human population grows, the amount of waste we produce also increases, creating serious problems for the environment. But what if these piles of trash could become something useful? One surprising answer is generating electricity from trash. This idea is helping countries around the world reduce waste while producing clean, reliable energy.

Generating electricity from trash means using waste that cannot be recycled to produce energy. Instead of sending all rubbish to landfills, special facilities collect and process it in ways that allow electricity to be created. Moreover, different types of waste can be used, including food waste, paper, wood, and other every day materials. So how do we turn waste into energy?

The basic idea is quite simple. Non-recyclable materials are placed in a special furnace where they are burned at very high temperatures. The heat produced is then used to boil water, creating steam. This steam spins large turbines connected to generators, which produce electricity. Additionally, there is another method which uses food waste and other natural materials. They get placed in large sealed containers where they normally break down and release a gas, methane, which is instantly collected and used to generate electricity.

By using waste to create electricity, it reduces the amount of trash sent to landfills and less rubbish ends up buried.

As the world's population continues to grow, finding smarter ways to manage waste becomes increasingly important. Generating electricity from trash offers an inventive solution that benefits both the people and the planet. It reduces landfill waste, provides a valuable source of energy, helps lower harmful gas emissions, and encourages a more sustainable way of thinking about the resources we use every day. While it is not the only answer to our environmental challenges, it is an important step towards a cleaner and greener future. We can transform one of society's biggest problems into one of its greatest opportunities.

Question V: Can anybody predict the timing of earthquakes accurately?

– *Arman Kani, 14*

Earthquakes are one of the most dangerous natural disasters in the world. They can destroy buildings, roads, bridges, and many people can lose their lives. Because of this, scientists have been trying to find a way to predict earthquakes before they happen.

Today, scientists can tell which places are more likely to have earthquakes by studying fault lines and the movement of tectonic plates. They also use special machines to record small movements under the ground. However, they still cannot predict the exact time, place, or strength of an earthquake.

Some people believe that animals can sense earthquakes before they happen. Others think that small earthquakes or changes in underground water may be warning signs. However, these ideas are not always correct, so scientists cannot depend on them.

Although earthquakes cannot be predicted accurately, many countries have early warning systems. These systems can detect the first earthquake waves and send alerts a few seconds before the stronger quakes start. Even a few seconds can help people move to a safer place.

In addition, advanced technology like satellites and artificial intelligence is now being used to study patterns in ground movement. Researchers use these modern tools to collect data and search for hidden signals before large quakes strike. While these technologies show great promise for the future, they are still in the testing stages.

In conclusion, predicting earthquakes is still very difficult. Scientists are doing more research and improving technology, but there is still no way to know exactly when an earthquake will happen. Until then, the best way to reduce the damage is to build stronger buildings, prepare for emergencies, and teach people what to do during an earthquake.

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mission to help the Iraqi youth where there
has been no help as yet!

CONCLUSION

We hope to further YSJOI's mission to empower the youth of Iraq in STEM fields by our monthly magazine, and sincerely hope we were able to help you and other students with this monthly issue! We are always open to feedback and questions, and everyone is of course welcome to learn in their own way; after all, we are all different, and we all belong.

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